DEEP LEARNING — 2026

YOLOv9

Learning What You Want to Learn

Using Programmable Gradient Information

GROUP MEMBERS

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VENUE

arXiv:2402.13616
Computer Vision & PR

DOMAIN

Real-Time Object Detection
MS COCO Benchmark

WHAT IS IT?

- PGI — Programmable Gradient Information: fixes unreliable gradients from information loss in deep layers
- GELAN — Generalized Efficient Layer Aggregation Network: flexible, lightweight, high-accuracy architecture

55.6%

AP on MS COCO
(YOLOv9-E)

PROBLEM STATEMENT & DATASET

THE CORE PROBLEM

Information Bottleneck

$$I(X,X) \geq I(X, f(X)) \geq I(X, g(f(X)))$$

Information is progressively lost across layers

CONSEQUENCES

- Loss of task-critical features during feedforward
- Unreliable gradients → incorrect learned correlations
- Poor convergence and unreliable gradients, especially in lightweight models.

EXISTING SOLUTIONS & THEIR LIMITS

Method	Limitation
Reversible Networks	Heavy inference cost (+20%)
Masked Modeling	Conflicts with target loss
Deep Supervision	Only works for very deep nets

→ Existing methods either hurt speed, create conflicting objectives, or are restricted to very deep architectures.

DATASET & TRAINING SETUP

MS COCO 2017 — Object Detection

- Train-from-scratch strategy • 500 epochs
- SGD optimizer • LR 0.01 → 0.0001 (linear decay)
- Linear warm-up (3 epochs) • Mosaic off (last 15)

ARCHITECTURE & TECHNICAL DETAILS

Programmable Gradient Information (PGI)

3 Components:

1. **Main Branch**

   Used for inference only — zero added deployment cost

2. **Auxiliary Reversible Branch**

   Training-only; generates reliable gradients by maintaining complete information

3. **Multi-level Auxiliary Information**

   Aggregates gradients across feature pyramid levels; prevents broken information

Insight: PGI doesn't force reversibility — it ensures gradients are reliable.

Generalized Efficient Layer Aggregation Network (GELAN)

Design foundation:

$$CSPNet + ELAN \rightarrow GELAN$$

- **Flexible Blocks:**

  Supports any computational block (Conv, Res, Dark, CSP)

- **Standard Convolutions:**

  Achieves better param efficiency than depth-wise designs like YOLO-MS

- **Hardware Agnostic:**

  Tune depth & blocks for any inference device

- **Stable Performance:**

  Stable scaling of parameters, computation, and accuracy.

★ **CSP-ELAN chosen: reduces params & improves AP +0.7%**

FUTURE WORK — PROJECT PHASE 2 PLAN

"Experimental evaluation of Programmable Gradient Information (PGI) under different training conditions."

M1 PGI Ablation Study

Core Contribution

OBJECTIVE

Verify PGI's main claim: train GELAN (no PGI) vs. YOLOv9 (with PGI)

METRICS

mAP • Training Loss • Convergence Speed

DATASET

Pascal VOC / COCO subset

★ YOLOv9 converges faster & achieves higher accuracy

M2 Training Dynamics

Learning Analysis

OBJECTIVE

Analyze how PGI changes learning behaviour during training

METRICS

Gradient Norm • Loss Curves • Validation mAP

DATASET

Same split as M1

★ Smoother loss & stable gradients with PGI

M3 Small Data Generalization

Robustness Study

OBJECTIVE

Test PGI under limited data: 100% / 50% / 20% dataset splits

METRICS

mAP across data fractions

DATASET

Stratified COCO / VOC subsets

★ YOLOv9 degrades less — PGI retains key information

Goal: Systematically evaluate how PGI affects model learning, convergence stability, and robustness under limited data.

CONCLUSION & Q&A

KEY CONTRIBUTIONS

PGI

Solves information bottleneck — reliable gradients for ALL model sizes, not just deep ones

GELAN

Efficient, flexible architecture — better param utilization than depth-wise convolution

YOLOv9

Beats ImageNet-pretrained models while training from scratch

KEY RESULTS ON MS COCO

YOLOv9 vs.	Params ↓	FLOPs ↓	AP ↑
YOLOv7 AF	42%	22%	Same 53%
YOLOv8-X	16%	27%	+1.7%
RT-DETR-X*	34%	19%	+0.8%

** RT-DETR-X uses ImageNet pretraining. YOLOv9 trains from scratch — a key advantage highlighted by the paper.*

TAKEAWAY

Better gradient design + efficient architecture
= SOTA detection without large pretrained datasets.

Questions & Answers — Thank You!